Efficient Large-Scale Learning of Minimax Risk Classifiers

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***Abstract*—Supervised learning with large-scale data usually leads to complex optimization problems, especially for classification tasks with multiple classes. Stochastic subgradient methods can enable efficient learning with a large number of samples for classification techniques that minimize the average loss over the training samples. However, recent techniques, such as minimax risk classifiers (MRCs), minimize the maximum expected loss and are not amenable to stochastic subgradient methods. In this paper, we present a learning algorithm based on the combination of constraint and column generation that enables efficient learning of MRCs with large-scale data for classification tasks with multiple classes. Experiments on multiple benchmark datasets show that the proposed algorithm provides upto a 10x speedup for general large-scale data and around a 100x speedup with a sizeable number of classes.**

***Index Terms*—Large-scale learning, multi-class classification, robust risk minimization, minimax risk classifiers, constraint generation**

I. Introduction

Large-scale data is common in multiple machine learning tasks, such as classification of handwritten digits, or sentiment analysis of text reviews [Akata et al., 2013, Wang et al., 2020]. Such data is composed by a large amount of training samples that are often represented as a high-dimensional vector of features [Li et al., 2010]. Supervised learning with large-scale data usually leads to complex optimization problems specially for classification tasks with multiple classes Bottou and Bousquet [2007], Bengio et al. [2010], Wang et al. [2020].

Stochastic subgradient (SS) methods can enable efficient learning for a large number of samples [Shalev-Shwartz et al., 2007, Bottou, 2010, Yuan et al., 2012, Kawaguchi and Lu, 2020]. These methods are amenable for classification techniques based on the empirical risk minimization (ERM) approach [Mohri et al., 2018] that minimizes the average risk over the training samples. Recent techniques, such as minimax risk classifiers (MRCs) [Mazuelas et al., 2020, 2023], are based on the robust risk minimization (RRM) approach that minimizes the maximum expected loss over the training samples and can provide performance guarantees in terms of worst-case error probability. Such techniques cannot leverage SS since the objective function is given by a maximization instead of an average over samples.

Efficient learning with a large number of features is often achieved by exploiting the parameters' sparsity induced by regularization methods [Singer and Duchi, 2009, Tan et al., 2010, Li et al., 2010, Mignacco et al., 2020, Celentano and Montanari, 2022]. In particular, recent methods have leveraged the parameters' sparsity using constraint generation approaches for linear optimization [Dedieu et al., 2022, Bondugula et al., 2023]. Specifically, the methods in Dedieu et al. [2022] have enabled the efficient learning of support vector machines (SVMs) with L1-penalization for a large number of features and/or instances. In addition, the methods in Bondugula et al. [2023] have enabled the efficient learning of MRCs with L1-penalization for a large number of features. However, the methods in Dedieu et al. [2022] are proposed for binary classification problems and those in Bondugula et al. [2023] only provide efficient learning with a small number of samples and classes. Specifically, the complexity per iteration in Bondugula et al. [2023] increases cubically with the number of samples and exponentially with the number of classes.

In this paper, we present a learning algorithm for MRCs that enables efficient learning with large-scale data and multiple classes. Specifically, the main contributions of the paper are as follows.

- We present an algorithmic framework based on the combination of constraint and column generation methods that enables the efficient learning of MRCs in general large-scale scenarios.
- The presented algorithm utilizes a greedy approach for constraint selection at each iteration that results in a complexity that scale quasi-linearly with the number of classes.
- For multiple scenarios of large-scale data, we present theoretical results that describe the convergence of the algorithm proposed.
- Using multiple benchmark datasets, we experimentally show that the proposed algorithm enables efficient learning of MRCs with large-scale data and multiple labels.

Notations: For a set $\mathcal{S}$, we denote its cardinality as $|\mathcal{S}|$; bold lowercase and uppercase letters represent vectors and matrices, respectively; for a matrix $\mathbf{F}$, and set of indices $\mathcal{I}$ and $\mathcal{J}$, $\mathbf{F}_{\mathcal{I},\mathcal{J}}$ denotes the submatrix obtained by the rows and columns corresponding to the indices in $\mathcal{I}$ and $\mathcal{J}$, respectively; for a vector $\mathbf{b}$ and set of indices $\mathcal{I}$, $\mathbf{b}_{\mathcal{I}}$ denotes the subvector obtained by the components corresponding to the indices in $\mathcal{I}$; for a vector $\mathbf{b}$ and an index i, b_i denotes the component

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at index i; $\mathbf{I}$ denotes the identity matrix; $\mathbb{1}\{\cdot\}$ denotes the indicator function; $\mathbf{1}$ denotes a vector of ones; for a vector $\mathbf{v}$, $|\mathbf{v}|$ and $(\mathbf{v})_+$, denote its component-wise absolute value and positive part, respectively; $\|\cdot\|_1$ denotes the 1-norm of its argument; $\otimes$ denotes the Kronecker product; $\preceq$ and $\succeq$ denote vector inequalities; $\mathbb{E}_{\mathrm{p}}\{\cdot\}$ denotes the expectation of its argument with respect to distribution p; and $\mathbf{e}_i$ denotes the i-th vector in a standard basis. For an integer a, we use $[a]$ to denote the set $\{1, 2, \ldots, a\}$.

II. Preliminaries and related work

In this section, we describe the setting addressed in the paper, and the MRC methods that minimize the worst-case error probability.

A. Problem formulation

Supervised classification uses sample-label pairs to determine classification rules that assign labels to samples. We denote by $\mathcal{X}$ and $\mathcal{Y}$ the sets of samples and labels, respectively, with $\mathcal{Y}$ represented by $\{1, 2, \ldots, |\mathcal{Y}|\}$. We denote by $\Delta(\mathcal{X}\times\mathcal{Y})$ the set of probability distributions on $\mathcal{X} \times \mathcal{Y}$ and by $\ell(\mathrm{h}, \mathrm{p})$ the expected 0-1 loss of the classification rule h with respect to distribution $\mathrm{p} \in \Delta(\mathcal{X} \times \mathcal{Y})$, i.e., $\ell(\mathrm{h}, \mathrm{p}) = \mathbb{E}_{\mathrm{p}}\mathbb{1}\{\mathrm{h}(x) \neq y\}$.

Sample-label pairs can be embedded into real-valued vectors using a feature map $\Phi : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}^m$. The most common approach to construct such a mapping involves combining multiple features over the samples with one-hot encodings of the labels as follows (see e.g., Mohri et al. [2018])

$$\Phi(x, y) = \left[\mathbb{1}\{y = 1\}\Psi(x)^{\mathrm{T}}, \ldots, \mathbb{1}\{y = |\mathcal{Y}|\}\Psi(x)^{\mathrm{T}}\right]^{\mathrm{T}}$$

where the map $\Psi : \mathcal{X} \to \mathbb{R}^d$ represents samples as real vectors of size d, for example, using random Fourier features [Rahimi and Recht, 2008].

In this paper we consider scenarios in which the number of samples n, number of classes $|\mathcal{Y}|$, and features' dimensionality $m = d|\mathcal{Y}|$ are large, and propose efficient learning methods for the MRCs, which are briefly described in the following.

B. Minimax Risk Classifiers

Minimax risk classifiers (MRCs) are classification rules that minimize the worst-case error probability over distributions in an uncertainty set [Mazuelas et al., 2020, 2022, 2023]. Specifically, such rules are solutions to the minimax risk problem defined as

$$\mathrm{R}^* = \min_{\mathrm{h}} \max_{\mathrm{p}\in\mathcal{U}} \ell(\mathrm{h}, \mathrm{p}) \tag{1}$$

where R^* is the worst-case error probability and $\mathcal{U}$ is an uncertainty set of distributions determined by expectation estimates as

$$\mathcal{U} = \{\mathrm{p} \in \Delta(\mathcal{X} \times \mathcal{Y}) : |\mathbb{E}_{\mathrm{p}}\{\Phi(x, y)\} - \boldsymbol{\tau}| \preceq \boldsymbol{\lambda}\}. \tag{2}$$

The mean vector $\boldsymbol{\tau}$ denotes expectation estimates corresponding with the feature mapping $\Phi : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}^m$, and the confidence vector $\boldsymbol{\lambda} \succeq \mathbf{0}$ accounts for inaccuracies in the estimate. The mean and confidence vectors can be obtained from the training samples $\{(x_i, y_i)\}_{i=1}^n$ as $\boldsymbol{\tau} = \frac{1}{n}\sum_{i=1}^n \Phi(x_i, y_i)$, $\boldsymbol{\lambda} = \lambda_0\mathbf{s}$, where $\mathbf{s}$ denotes the vector formed by the component-wise sample standard deviations of $\{\Phi(x_i, y_i)\}_{i=1}^n$ and λ_0 is a regularization parameter.

As described in Mazuelas et al. [2020, 2022, 2023] using the expected 0-1 loss, the MRC rule solution of (1) is given by a linear combination of the feature mapping. Specifically, the minimax rule assigns labels as $\arg\max_{y\in\mathcal{Y}} \Phi(x, y)^{\mathrm{T}}\boldsymbol{\mu}^*$. The vector $\boldsymbol{\mu}^*$ that determines the 0-1 MRC rule corresponding to (1) is obtained by solving the convex optimization problem [Mazuelas et al., 2020, 2023]

$$\min_{\boldsymbol{\mu}\in\mathbb{R}^m} 1 - \boldsymbol{\tau}^{\mathrm{T}}\boldsymbol{\mu} + \varphi(\boldsymbol{\mu}) + \boldsymbol{\lambda}^{\mathrm{T}}|\boldsymbol{\mu}|, \tag{3}$$

with $\varphi(\boldsymbol{\mu}) = \max_{x\in\mathcal{X},\mathcal{C}\subseteq\mathcal{Y}} (\sum_{y\in\mathcal{C}} \Phi(x, y)^{\mathrm{T}}\boldsymbol{\mu} - 1)/|\mathcal{C}|$. In addition, the minimum value of (3) equals the minimax risk value of (1). The minimax risk value provides an upper bound on the expected classification error when the underlying distribution of the training samples is included in the uncertainty set.

The convex optimization problem (3) for MRCs includes an L1-penalty $\boldsymbol{\lambda}^{\mathrm{T}}|\boldsymbol{\mu}|$, which induces sparsity in the coefficients $\boldsymbol{\mu}^*$ associated with the feature mapping. This sparsity indicates that only a subset of features is necessary to achieve the optimal worst-case error probability. Consequently, efficient learning can be realized by focusing on the most relevant features when solving the MRC optimization problem.

C. LP formulation of MRCs

The convex optimization problem (3) of 0-1 MRCs can be formulated as the following linear program (LP) [Bondugula et al., 2023]

$$\begin{aligned} \mathcal{P} : \min_{\boldsymbol{\mu}_1,\boldsymbol{\mu}_2,\nu} \; & -(\boldsymbol{\tau} - \boldsymbol{\lambda})^{\mathrm{T}}\boldsymbol{\mu}_1 + (\boldsymbol{\tau} + \boldsymbol{\lambda})^{\mathrm{T}}\boldsymbol{\mu}_2 + \nu \\ \text{s.t.} \; & \mathbf{F}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) - \nu\mathbf{1} \preceq \mathbf{b}, \; \boldsymbol{\mu}_1, \boldsymbol{\mu}_2 \succeq 0. \end{aligned} \tag{4}$$

The matrix $\mathbf{F}$ and vector $\mathbf{b}$ that define the constraints can be represented as

$$\mathbf{F} = \begin{bmatrix} \mathbf{F}^{x_1} \\ \mathbf{F}^{x_2} \\ \vdots \\ \mathbf{F}^{x_n} \end{bmatrix}, \; \mathbf{b} = \begin{bmatrix} \hat{\mathbf{b}} \\ \hat{\mathbf{b}} \\ \vdots \\ \hat{\mathbf{b}} \end{bmatrix} \tag{5}$$

such that a pair of submatrix $\mathbf{F}^{x_i}$ and subvector $\hat{\mathbf{b}}$ defines all the constraints corresponding to each training instance x_i, for $i = 1, 2, \ldots, n$. Each row of the submatrix $\mathbf{F}^{x_i}$ and component of subvector $\hat{\mathbf{b}}$ corresponds to a non-empty subset $\mathcal{C} \subseteq \mathcal{Y}$, and is defined as $\sum_{y\in\mathcal{C}} \Phi(x_i, y)^{\mathrm{T}}/|\mathcal{C}|$ and $1/|\mathcal{C}| - 1$, respectively, for a given $x_i \in \mathcal{X}$. The number of variables $q = 2m + 1$ in (4) is given by the number of features m while the number of constraints $p = n(2^{|\mathcal{Y}|} - 1)$ in (4) is given by the number of training samples n and the non-empty subsets of $\mathcal{Y}$. Optimization problem (3) can be solved by addressing the LP in (4) at learning because the optimal values of (3) and (4) coincide, and a solution of (3) $\boldsymbol{\mu}^*$ can be obtained as $\boldsymbol{\mu}^* = \boldsymbol{\mu}_1^* - \boldsymbol{\mu}_2^*$ for $\boldsymbol{\mu}_1^*$ and $\boldsymbol{\mu}_2^*$ solution of (4).

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The lagrange dual of (4) is given by

$$\begin{aligned} \mathcal{D}: \max_{\boldsymbol{\alpha}} \quad & -\mathbf{b}^{\mathrm{T}}\boldsymbol{\alpha} \\ \text{s.t. } & \boldsymbol{\tau}-\boldsymbol{\lambda} \preceq \mathbf{F}^{\mathrm{T}}\boldsymbol{\alpha} \preceq \boldsymbol{\tau}+\boldsymbol{\lambda} \\ & \mathbf{1}^{\mathrm{T}}\boldsymbol{\alpha}=1,\ \boldsymbol{\alpha}\succeq 0. \end{aligned} \tag{6}$$

The number of variables in the dual LP in (6) is equal to the number of constraints in the primal, that is, p, while the number of constraints in (6) is equal to the number of variables in the primal, that is, q.

The complexity of MRC learning given by such LP formulation is not affordable when either the number of samples n, the number of features m, or the number of classes $|\mathcal{Y}|$ are large. In particular, the number of variables and constraints in (4) grow linearly with m and n, respectively, while the number of constraints in (4) grows exponentially with $|\mathcal{Y}|$. In the following, we describe the general constraint generation methodology that we use in the remaining of the paper to provide an efficient learning algorithm for large-scale MRCs.

D. Constraint Generation for Linear Programming

Constraint generation methods can enable to efficiently solve large-scale LPs when the number of constraints is large [Bertsimas and Tsitsiklis, 1997, Desrosiers and Lübbecke, 2005]. The basic idea is to start with a candidate set of constraints and iteratively add new constraints until the resulting solution is feasible for the whole problem. Such procedure is effective when the number of variables is significantly smaller than the number of constraints since only a small set of constraints will be active in the solution. Moreover, this procedure can also be used for large number of variables by considering the corresponding dual LP. Since generating (adding) constraints in the dual is equivalent to adding variables in the primal, such method is also known as column generation. In the following, we use constraint and column generation tools to provide a learning algorithm for the LP formulation of MRCs that is efficient for large-scale data and multi-class classification.

III. Efficient Large-Scale Learning of MRCs

In this section, we propose a learning algorithm based on the combination of constraint and column generation for the LP formulation of 0-1 MRCs. We present implementation details along with theoretical guarantees for multiple scenarios such as large number of samples but few features, and large number of samples with many features. In addition, we describe efficient techniques for learning with multiple classes.

A. General Algorithm for Efficient Learning

In the following, we present an iterative algorithm based on the combination of constraint and column generation that efficiently solves the 0-1 MRCs LP (4) (see Algorithm 1 for the pseudocode). Efficient learning for MRCs can be achieved since only a small subset of variables and constraints are active at the solution of the LP formulation for multiple settings of large-scale data, as detailed in Sections III-B and III-C.

The proposed Algorithm 1 obtains the optimal solution $\boldsymbol{\mu}^*$ by iteratively solving a sequence of subproblems.

Algorithm 1 Efficient learning of MRCs for large-scale data

Input: $\mathbf{F}, \mathbf{b}, \boldsymbol{\tau}$, and $\boldsymbol{\lambda}$,
initial subset of features $\mathcal{J}$ and constraints $\mathcal{I}$,
initial solution $\boldsymbol{\mu}_1^1$, $\boldsymbol{\mu}_2^1$, ν^1 of $\mathcal{P}_{\mathcal{I},\mathcal{J}}$,
initial solution $\boldsymbol{\alpha}^1$ of $\mathcal{D}_{\mathcal{I},\mathcal{J}}$,
primal constraints' violation threshold ϵ_1,
dual constraints' violation threshold ϵ_2,
constraint limit per iteration n_{max},
feature limit per iteration m_{max}.
maximum number of iterations k_{max}.
Output: optimal solution $\boldsymbol{\mu}^* \in \mathbb{R}^m$,
worst-case error probability R^*.

$k \leftarrow 1$
repeat
  $\hat{\mathcal{J}} \leftarrow \mathcal{J},\ \hat{\mathcal{I}} \leftarrow \mathcal{I}$
  $\mathcal{J} \leftarrow \text{FEAT}(\hat{\mathcal{I}},\ \hat{\mathcal{J}},\ \boldsymbol{\alpha}^k,\ \epsilon_2,\ m_{\text{max}})$
  $\mathcal{I} \leftarrow \text{CONSTR}(\hat{\mathcal{I}},\ \hat{\mathcal{J}},\ \boldsymbol{\mu}_1^k,\ \boldsymbol{\mu}_2^k,\ \nu^k,\ \epsilon_1,\ n_{\text{max}})$
  LPSOLVE$\big(\mathcal{I}, \mathcal{J}\big)$
  $\boldsymbol{\mu}_1^{k+1},\ \boldsymbol{\mu}_2^{k+1},\ \nu^{k+1} \leftarrow$ Solution of primal $\mathcal{P}_{\mathcal{I},\mathcal{J}}$
  $\boldsymbol{\alpha}^{k+1} \leftarrow$ Solution of dual $\mathcal{D}_{\mathcal{I},\mathcal{J}}$
  $\mathrm{R}^{k+1} \leftarrow$ Optimal value
  $k \leftarrow k+1$
until $\mathcal{I} \setminus \hat{\mathcal{I}} = \emptyset$ and $\mathcal{J} \setminus \hat{\mathcal{J}} = \emptyset$ and $k \leq k_{\text{max}}$
$\boldsymbol{\mu}^* = [0, 0, \ldots, 0] \in \mathbb{R}^m$
$\boldsymbol{\mu}^*_{\mathcal{J}} \leftarrow \boldsymbol{\mu}_1^k - \boldsymbol{\mu}_2^k,\ \mathrm{R}^* \leftarrow \mathrm{R}^k$

These subproblems correspond to the LP in (4) defined over a subset of constraints and features. Specifically, the subproblem corresponding to the subset of constraints $\mathcal{I} \subseteq \{1, 2, \ldots, p\}$ and features $\mathcal{J} \subseteq \{1, 2, \ldots, m\}$ is defined as

$$\begin{aligned} \mathcal{P}_{\mathcal{I},\mathcal{J}}: \min_{\boldsymbol{\mu}_1,\boldsymbol{\mu}_2,\nu} \quad & -(\boldsymbol{\tau}-\boldsymbol{\lambda})_{\mathcal{J}}^{\mathrm{T}}\boldsymbol{\mu}_1 + (\boldsymbol{\tau}+\boldsymbol{\lambda})_{\mathcal{J}}^{\mathrm{T}}\boldsymbol{\mu}_2 + \nu \\ \text{s.t. } & \mathbf{F}_{\mathcal{I},\mathcal{J}}(\boldsymbol{\mu}_1-\boldsymbol{\mu}_2) - \nu\mathbf{1} \preceq \mathbf{b}_{\mathcal{I}},\ \boldsymbol{\mu}_1, \boldsymbol{\mu}_2 \succeq 0. \end{aligned} \tag{7}$$

In addition, the dual of (7) is

$$\begin{aligned} \mathcal{D}_{\mathcal{I},\mathcal{J}}: \max_{\boldsymbol{\alpha}} \quad & -\mathbf{b}_{\mathcal{I}}^{\mathrm{T}}\boldsymbol{\alpha} \\ \text{s.t. } & (\boldsymbol{\tau}-\boldsymbol{\lambda})_{\mathcal{J}} \preceq (\mathbf{F}_{\mathcal{I},\mathcal{J}})^{\mathrm{T}}\boldsymbol{\alpha} \preceq (\boldsymbol{\tau}+\boldsymbol{\lambda})_{\mathcal{J}} \\ & \mathbf{1}^{\mathrm{T}}\boldsymbol{\alpha}=1, \boldsymbol{\alpha}\succeq 0. \end{aligned} \tag{8}$$

Each iteration k of Algorithm 1 solves subproblems (7) and (8), and obtains primal solution $\boldsymbol{\mu}^k$, ν^k and the dual solution $\boldsymbol{\alpha}^k$ along with the worst-case error probability R^k given by the optimal value. The primal solution is used by the function CONSTR to obtain the subsequent primal constraints based on the constraints' violation. The dual solution is used by the function FEAT to obtain the subsequent set of features based on the dual constraints' violation.

The function FEAT obtains the subsequent set of features $\mathcal{J}$ by adding and/or removing features based on the violations in the corresponding dual constraints. In particular, each feature $j \in [m]$ corresponds to two variables in the primal ${\mu_1}_j^k$ and ${\mu_2}_j^k$ that correspond to the two constraints $(\mathbf{F}_{\hat{\mathcal{I}},j})^{\mathrm{T}}\boldsymbol{\alpha}^k \geq \tau_j - \lambda_j$ and $(\mathbf{F}_{\hat{\mathcal{I}},j})^{\mathrm{T}}\boldsymbol{\alpha}^k \leq \tau_j + \lambda_j$ in the dual. Therefore, a new feature $j \in [m] \setminus \hat{\mathcal{J}}$ is added to the set $\mathcal{J}$ if one of the cor-

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responding dual constraints is violated by at least $\epsilon_2 \geq 0$, that is, $|(\mathbf{F}_{\hat{\mathcal{I}},j})^{\mathrm{T}}\boldsymbol{\alpha}^k - \tau_j| - \lambda_j \geq \epsilon_2$. In addition, an existing feature $j \in \mathcal{J}$ is removed if the corresponding dual constraints are overly-satisfied, that is, $|(\mathbf{F}_{\hat{\mathcal{I}},j})^{\mathrm{T}}\boldsymbol{\alpha}^k - \tau_j| - \lambda_j < 0$. Similarly, the function CONSTR obtains the subsequent set of primal constraints $\mathcal{I}$ by adding and/or removing constraints based on the violations. In particular, a new constraint $i \in [p] \setminus \hat{\mathcal{I}}$ is added to the set $\mathcal{I}$ if it is violated by at least $\epsilon_1 \geq 0$, that is, $\mathbf{F}_{i,\hat{\mathcal{J}}}(\boldsymbol{\mu}_1^k - \boldsymbol{\mu}_2^k) - \nu^k - \mathrm{b}_i \geq \epsilon_1$. In addition, an existing constraint $i \in \mathcal{I}$ is removed if it is overly-satisfied, that is, $\mathbf{F}_{i,\hat{\mathcal{J}}}(\boldsymbol{\mu}_1^k - \boldsymbol{\mu}_2^k) - \nu^k - \mathrm{b}_i < 0$.

The function FEAT adds $m_{\max}$ features corresponding to the $m_{\max}$ most violated constraints in the dual. This greedy selection process requires evaluating all the constraints in the dual so that its complexity is $O(q)$. On the other hand, the function CONSTR adds $n_{\max}$ violated constraints of the primal using the greedy algorithm described in the following to achieve a significantly lower complexity than $O(p)$.

1) Efficient evaluation of primal constraints violation: The number of constraints p in the primal MRC LP (4) is given by $n(2^{|\mathcal{Y}|}-1)$. Evaluating all the constraints to find the maximum violation has an exponential complexity in terms of the number of classes $|\mathcal{Y}|$ that is not affordable in multi-class settings. In the following, we present Algorithm 2 that computes the constraint with maximum violation in a significantly lower complexity.

Algorithm 2 Efficient evaluation of constraints violation

Input: instance $x \in \mathcal{X}$,
feature mapping Φ,
primal solution $\boldsymbol{\mu}$
Output: index i^* for the subset
corresponding to maximum in $\varphi(\boldsymbol{\mu})$

1: $\mathbf{v} = [\Phi(x,1), \Phi(x,2), \ldots, \Phi(x,|\mathcal{Y}|)]^{\mathrm{T}}\boldsymbol{\mu}$
2: $i_1, i_2, \ldots, i_{|\mathcal{Y}|}$ = ARGSORT($\mathbf{v}$)
3: $\mathbf{c} = [0, 0, \ldots, 0] \in \mathbb{R}^{|\mathcal{Y}|}$
4: $\psi = \mathrm{v}_{i_1} - 1, \mathrm{c}_{i_1} = 1$
5: **for** $k = 2, 3, \ldots, |\mathcal{Y}|$ **do**
6: $\quad\hat{\psi} = ((k-1)\psi + \mathrm{v}_{i_k})/k$
7: $\quad$**if** $\hat{\psi} \geq \psi$ **then**
8: $\quad\quad\psi = \hat{\psi}$, $\mathrm{c}_{i_k} = 1$
9: $\quad$**end if**
10: **end for**
11: $i^* = \sum_{j=1}^{|\mathcal{Y}|} \mathrm{c}_j 2^{j-1}$

The MRC LP (4) has exponential number of constraints corresponding to each instance $x_i \in \mathcal{X}$. The exponential number of constraints correspond to different non-empty subsets $\mathcal{C} \subseteq \mathcal{Y}$ as defined by the rows and components of submatrix $\mathbf{F}^{x_i}$ and subvector $\hat{\mathbf{b}}$ in (5) for $x_i \in \mathcal{X}$. The Algorithm 2 returns the index $i^* \in \{1, 2 \ldots, 2^{|\mathcal{Y}|} - 1\}$ corresponding with the subset achieving the maximum constraint. Specifically, a subset achieving the maximum constraint corresponds to the subset achieving the maximum value

$$\psi = \max_{\mathcal{C} \subseteq \mathcal{Y}} \frac{\sum_{y \in \mathcal{C}} \Phi(x_i, y)^{\mathrm{T}}\boldsymbol{\mu} - 1}{|\mathcal{C}|} \tag{9}$$

for an $x_i \in \mathcal{X}$ and given solution $\boldsymbol{\mu}$. The Algorithm 2 computes the maximum value ψ over all subsets in a greedy fashion similar to the approach presented in Fathony et al. [2016]. In particular, the algorithm computes $\Phi(x_i, y)^{\mathrm{T}}\boldsymbol{\mu}$ corresponding to each $y \in \mathcal{Y}$ and sorts them in decreasing order. Then, the algorithm starts with empty set $\mathcal{C} = \emptyset$ and iteratively adds a label to the subset $\mathcal{C}$ in the sorted order until adding a label does not increase the value of $\hat{\psi}$. Such greedy approach obtains the maximum value ψ over all the subsets since the maximum value for a fixed subset length is obtained by the set with labels corresponding with maximum values of $\Phi(x_i, y)^{\mathrm{T}}\boldsymbol{\mu}$. Therefore, the computational complexity of Algorithm 1 is $O(|\mathcal{Y}| \log |\mathcal{Y}|)$ due to the sorting in line 2.

In the following, we present the implementation details along with the theoretical properties of the proposed learning algorithm for multiple scenarios.

B. Learning with a large number of samples

A large number of samples together with a reduced number of features leads to an MRC primal LP with a large number of constraints and few variables. In such scenario, usually a small set of constraints will be active in the solution Bertsimas and Tsitsiklis [1997]. Algorithm 1 enables efficient learning by iteratively selecting the relevant subset of constraints using the CONSTR function. In particular, such a function adds and removes constraints (as defined in Section III-A) along the iterations while considering all the features, that is, the function FEAT is disabled in this scenario.

The following theorem shows that the proposed algorithm for a large number of samples provides an increasing sequence of worst-case error probabilities and converges to the worst-case error probability given by the MRC LP corresponding with all the constraints.

Theorem 1. *Let* R^* *be the worst-case error probability obtained by solving* (4) *using all the constraints and features. If* $\mathrm{R}^k, k = 1, 2 \ldots,$ *is the sequence of optimal values obtained by adding and removing constraints along the iterations of the proposed algorithm. Then, we have*

$$\mathrm{R}^k \leq \mathrm{R}^{k+1}. \tag{10}$$

Moreover, if $\hat{\epsilon}_1$ *is the largest violation in the constraints of the primal at iteration* k_0*, then at any iteration* $k \geq k_0$*,*

$$\mathrm{R}^* - \hat{\epsilon}_1 \leq \mathrm{R}^k \leq \mathrm{R}^*. \tag{11}$$

Proof. See Appendix A. □

Inequality (10) shows that the algorithm obtains an increasing sequence of worst-case error probabilities along the iterations. On the other hand, the inequality (11) shows that the algorithm finds the MRC corresponding to all the constraints when there is no violation in the primal constraints, that is,

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$\hat{\epsilon}_1 = 0$. Such a case occurs after a finite number of iterations due to the properties of constraint generation methods Bertsimas and Tsitsiklis [1997]. In other cases, it finds an approximate solution with accuracy that depends directly on the largest violation $\hat{\epsilon}_1$. Moreover, if the algorithm terminates with $k < k_{\max}$, then the hyper-parameter ϵ_1 is the largest violation which can provide a trade-off between the accuracy and the complexity of the algorithm (larger values for ϵ_1 decrease the number of iterations but lead to approximate solutions). In the following, we present the implementation details for the scenario with large number of samples and features along with the corresponding theoretical analysis.

C. Learning with a large number of samples and features

A large number of samples and features corresponds to large number of constraints and variables in the MRC primal LP. Efficient learning can be achieved in such scenarios leveraging the sparsity in the MRCs solution due to the L1-penalization. In particular, the solution sparsity implies that MRCs learning can be carried out using a small subset of features, that is, a small subset of variables in the MRC LP. Moreover, a small subset of variables implies that only a subset of constraints will be active in the solution. Algorithm 1 enables efficient learning by iteratively selecting the relevant variables and constraints. In particular, Algorithm 1 generates the relevant subset of features and constraints using the functions FEAT and CONSTR that add features and constraints (as defined in Section III-A) along the iterations. In this scenario, functions CONSTR and FEAT do not remove constraints or features to ensure a proper convergence of the algorithm.

The following theorem bounds the worst-case error probability obtained by the proposed algorithm at any iteration. In addition, such result shows that the worst-case error probability converges to the worst-case error probability of the MRCs LP corresponding with all the constraints and features.

Theorem 2. *Let $\boldsymbol{\mu}^*$ and R^* be the MRC coefficient and the worst-case error probability obtained by solving* (4) *using all the constraints and features. If R^k is the worst-case error probability obtained at iteration k of Algorithm 1. Then,*

$$\mathrm{R}^* - \hat{\epsilon}_1 \leq \mathrm{R}^k \leq \mathrm{R}^* + \hat{\epsilon}_2 \|\boldsymbol{\mu}^*\|_1 \tag{12}$$

for the largest violations $\hat{\epsilon}_1$ and $\hat{\epsilon}_2$ in the constraints of the primal and dual at iteration k.

Proof. See Appendix B. □

Inequality (12) shows that the algorithm finds an MRC corresponding with all the features and constraints when there is no violation, that is, $\hat{\epsilon}_1 = 0$ and $\hat{\epsilon}_2 = 0$. Such a case occurs after a finite number of iterations as the algorithm adds multiple constraints and features in each iteration. In case $\hat{\epsilon}_1 > 0$ and $\hat{\epsilon}_2 = 0$, the algorithm can under-estimate the worst-case error while in case of $\hat{\epsilon}_1 = 0$ and $\hat{\epsilon}_2 > 0$, the algorithm can over-estimate the worst-case error. In other cases, it obtains an approximate solution that depends on the largest violations $\hat{\epsilon}_1$ and $\hat{\epsilon}_2$. Furthermore, if the algorithm terminates with $k < k_{\max}$, then the hyper-parameters ϵ_1 and ϵ_2 are the largest violations and can provide a trade-off between training time and the optimality of the solution.

D. Computational complexity

The computational complexity of Algorithm 1 is determined by two main factors: the number of iterations and the cost per iteration. These, in turn, depend on the maximum number of features and constraints selected in each iteration, denoted by $m_{\max}$ and $n_{\max}$, as well as the constraint violation thresholds ϵ_1 and ϵ_2. Reducing $n_{\max}$ or $m_{\max}$ lowers the per-iteration cost but typically increases the number of iterations required. Conversely, increasing ϵ_1 or ϵ_2 can reduce both the number of iterations and the per-iteration complexity at the cost of obtaining approximate solutions. In addition to the hyper-parameters $n_{\max}$, $m_{\max}$, ϵ_1, and ϵ_2, the per-iteration complexity also depends on the total number of features m, the number of samples n, and the number of classes $|\mathcal{Y}|$. Specifically, the functions CONSTR and FEAT have time complexities of $O(n|\mathcal{Y}|\log|\mathcal{Y}|)$ and $O(m \log m)$, respectively. The complexity of CONSTR function arises due to a scan over all samples using Algorithm 2, while the complexity of FEAT function arises from sorting operations required to identify features associated with the most violated dual constraints (see Section III-A for details).

The number of iterations can be further minimized by carefully selecting the initial subsets of constraints and features. Additionally, choosing an appropriate initial set of constraints is essential to ensure that the feasible region of the MRC LP is bounded from the very first iteration. A common strategy in constraint generation methods involves using the solution obtained from running a few iterations of a first-order optimization method as a basis for constructing these initial subsets [Dedieu et al., 2022]. However, the computational cost of first order methods for MRCs can be large with many samples. We propose to use a simple approach based on clustering to obtain a reasonable choice for the subset of constraints. On the other hand, the corresponding subset of features can be directly obtained using the method in Bondugula et al. [2023]. Specifically, a subset of constraints are obtained as that corresponding to $|\mathcal{Y}|$ samples representing the centers of clusters for different classes. In addition, the constraints corresponding to such samples ensure a bounded feasible region (see Section III-E for the details).

The complexity per iteration can further improved in case of large number of samples with few features by leveraging the warm-start in LPs. In such cases, the warm-start for iteration k is directly derived from the dual solution $\boldsymbol{\alpha}^{k-1}$ of the previous iteration $k-1$. The warm-start serves as a basic feasible solution for the dual LP at iteration k, as it is constructed by discarding the primal constraints with positive slack or the dual variables with zero coefficients from the solution at iteration $k-1$.

E. Initialization

An adequate choice for the initial subset of constraints and features can reduce the number of iterations for termination of Algorithm 1. Moreover, an adequate subset of constraints

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is required to ensure a bounded feasible region for the initial LP at the first iteration of Algorithm 1. In the following, we present an efficient approach based on clustering to obtain an appropriate initial subset of constraints $\mathcal{I}$ and avoid the unbounded feasible region for the initial LP. Note that the corresponding initial subset of features $\mathcal{J}$ can be directly obtained by a few iterations of the learning algorithm presented in Bondugula et al. [2023] on the LP defined by $\mathcal{I}$.

Consider the following $|\mathcal{Y}|$ samples corresponding to the center of the clusters for each class.

$$\hat{\mathcal{X}} = \{\hat{\boldsymbol{x}}_i : \hat{\boldsymbol{x}}_i = \boldsymbol{\tau}^i / \mathrm{p}_i^y \ \forall \ i = 1, 2, \ldots, |\mathcal{Y}|\} \tag{13}$$

where $\boldsymbol{\tau}^i \in \mathbb{R}^d$ denotes the components of $\boldsymbol{\tau}$ corresponding to class i and $\mathbf{p}^y \in \mathbb{R}^{|\mathcal{Y}|}$ is a vector of class proportions, that is, $\mathrm{p}_i^y = \sum_{j=0}^{n} \mathbb{1}\{y_j = i\}/n$ for n training samples. The MRC optimization problem (3) solved using samples in $\hat{\mathcal{X}}$, and the estimates $\boldsymbol{\tau}$ and $\boldsymbol{\lambda}$ leads to a bounded LP since the uncertainty set (2) is non-empty. Specifically, the $|\mathcal{Y}|$ samples ensure non-emptiness as a distribution with weights $\mathbf{p}^y$ is inside the set since $\sum_{i=0}^{|\mathcal{Y}|} \mathrm{p}_i^y \Phi(\hat{\boldsymbol{x}}_i, y_i) = \boldsymbol{\tau}$ for $\hat{\boldsymbol{x}}_i \in \hat{\mathcal{X}}$. Therefore, the initial MRCs LP (4) given by the set of constraints corresponding to the samples in $\hat{\mathcal{X}}$ is bounded.

The initial MRC LP using all the $|\mathcal{Y}|(2^{|\mathcal{Y}|} - 1)$ constraints defined over the $|\mathcal{Y}|$ samples in (13) ensures a bounded feasible region. Initializing Algorithm 1 using such constraints is a viable option upto a few labels. However, the initialization can be computational inefficient with a large number of classes as the number of constraints in the initial LP become significantly large. Therefore, we propose a variation of the above initialization that can be useful for a large number of classes. Specifically, we use a reduced subset from the constraints obtained from $\hat{\mathcal{X}}$ as initialization over an MRCs LP with an additional constraint enforcing the objective to be positive. Notice that the objective value of the MRC LP has to be a positive value since it is the worst-case error probability. Therefore, the additional constraint does not affect the optimal solution and ensures a feasible bounded region for the LP. The initial solution obtained for the reduced subset of constraints over the redefined MRC LP can be trivial, that is, zero. However, Algorithm 1 eventually obtains a non-trivial solution by iteratively selecting the constraints and features.

The proposed algorithm for MRCs provides efficient learning for multiple scenarios of large-scale learning. The theoretical analysis shows that the algorithm can deal with accuracy vs efficiency trade-offs by selecting appropriate values for the hyper-parameters ϵ_1 and ϵ_2. In the next section, we further assess the efficiency achieved through numerical experiments using multiple large-scale datasets.

IV. Results

In this section, we present experimental results that show comparison with state-of-the-art methods in terms of precision in the solution and training time. All the results are obtained using the hyper-parameters ϵ_1 =1e-2, ϵ_2 =1e-5, and $n_{\max} = m_{\max} = 400$ for the proposed algorithm (see Appendix C for results with other hyper-parameter settings). The experiments are performed in Python 3.9 with a memory limit of 180GB and a time limit of 2e+5 secs. In addition, we use Gurobi optimizer 9.5.2 in our experiments for solving LPs.

Table I. Datasets.

Dataset	Samples (n)	Features (d)	Classes ($\lvert\mathcal{Y}\rvert$)
pulsar	17898	8	2
house16	22784	16	2
cats vs dogs	23262	512	2
yearbook	37921	512	2
rcv1	20242	47236	2
real_sim	72309	20958	2
news20	19996	1355191	2
satellite	6435	36	6
dry_bean	13611	16	7
optdigits	5620	64	10
mnist	70000	512	10
fashion_mnist	70000	512	10
cifar10	60000	512	10
cifar100	60000	512	100

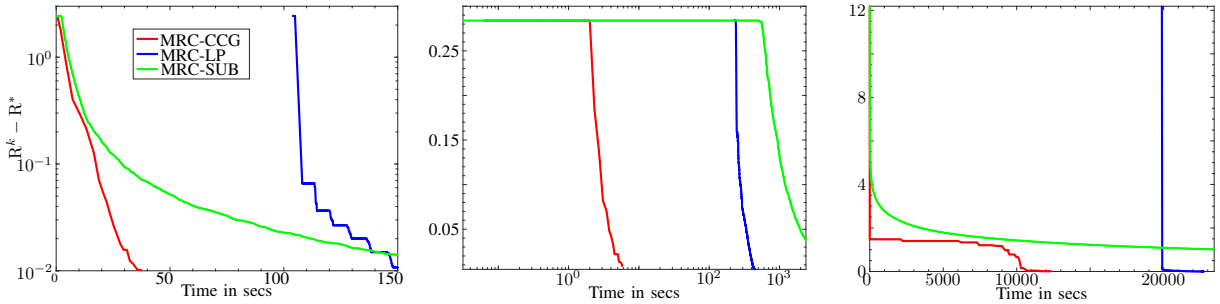
The proposed algorithm (MRC-CCG) is compared with the LP formulation of MRCs (4) using all constraints and variables (MRC-LP), and the accelerated subgradient method (MRC-SUB) presented in Mazuelas et al. [2023] which is implemented in the python library MRCpy [Bondugula et al., 2024]. In addition, the proposed algorithm is compared with constraint generation methods for L1-SVMs (SVM-CCG) presented in Dedieu et al. [2022] for binary classification and the L1-SVMs for multi-class classification (SVM-MULTI) presented in Wang et al. [2006]. We set the regularization parameter for all the methods to 0.01.

The experimental results are obtained using 7 binary and 7 multi-class datasets summarized in Table I. We use the ResNet18 [He et al., 2016] features for the image datasets 'cats vs dogs', 'yearbook', 'mnist', 'fashion_mnist', 'cifar10', and 'cifar100', and augment the features of the tabular datasets 'pulsar', 'house16', 'satellite', 'dry_bean', and 'optdigits' with 400 random Fourier features. The proposed algorithm (MRC-CCG) is implemented in the python library MRCpy [Bondugula et al., 2024].

A. Fast convergence and improved scalability

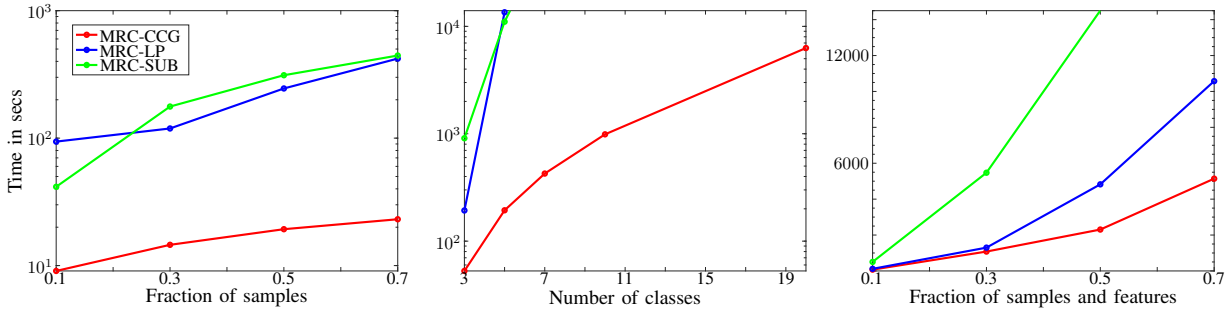
Figure 1 illustrates the fast convergence to R^* achieved by the proposed iterative algorithm MRC-CCG in comparison with the other algorithms MRC-SUB and MRC-LP. The worst-case error R^k is obtained along the iterations $k = 1, 2, \ldots$ of each algorithm and the optimal worst-case error R^* corresponds to the optimal worst-case error obtained by the MRC-LP that solves (4) using all the constraints and variables. Figure 1a and Figure 1b correspond with a binary and multi-class classification problems with a large number of samples while Figure 1c corresponds to a binary classification problem with a large number of samples and features. All the times are averaged over 10 repetitions using 70% of the data. The figures show that MRC-CCG converges to R^* significantly faster than the existing methods MRC-SUB and MRC-LP in all scenarios. Such improvement is even more clear for the

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(a) cats vs dogs dataset *(b)* satellite dataset *(c)* rcv1 dataset

Figure 1. Convergence of the worst-case error probability R^k over time. The figures correspond to different scenarios of large-scale learning and demonstrate that MRC-CCG achieves a fast convergence in comparison with state-of-the-art learning methods for MRCs.



(a) cats vs dogs dataset *(b)* cifar100 dataset *(c)* rcv1 dataset

Figure 2. Illustration of scalability for multiple scenarios of large-scale learning. The figures demonstrate that MRC-CCG achieves an improved scalability in comparison with state-of-the-art learning methods for MRCs.

multi-class scenario due to the complexity improvement from exponential to quasi-linear in the number of classes. Moreover, we observe that the difference in the worst-case error $R^k - R^*$ monotonically decreases with increasing time in case of large number of samples (Figure 1a and 1b) due to the properties shown in Theorem 1.

Figure 2 illustrates the improved scalability of MRC-CCG in comparison with the existing methods MRC-SUB and MRC-LP. Specifically, the figure shows the training times obtained for increasing the number of samples, the number of classes, and the number of both samples and features. All the training times are averaged over 10 random partitions of data. The figures show that MRC-CCG achieves a significantly more efficient learning of MRCs with increasing data and classes. In particular, Figure 2b shows that the algorithm achieves significant improvement in training times for multi-class setting and scales to a sizeable number of classes.

B. Training times and precision

Table II quantifies the improvement in training time and the precision in the objective value (in terms of average error (AE)) obtained by MRC-CCG with respect to MRC-LP and MRC-SUB using multiple binary classification datasets. All the results are averaged over 10 random repetitions using 80% of the data and the AE in the objective value is computed as the difference with respect to the objective value obtained by solving the LP (4) using all the constraints and variables. In addition, the table presents results for the constraint generation methods for L1-SVMs (SVM-CCG) and the LP formulation of L1-SVMs (SVM-LP). Notice that the large AE corresponding to SVMs represents the differences in large objective.

Table III quantifies the improvement in training time by MRC-CCG with respect to MRC-SUB, and the L1-SVMs for multi-class classification (SVM-MULTI) since the efficient learning method (SVM-CCG) for L1-SVM is proposed for binary classification problems [Dedieu et al., 2022]. Note that the empty cells in the table for SVM-LP, SVM-MULTI, MRC-LP, and MRC-SUB correspond to the cases where the method could not obtain a solution due to their requirements for computational resources.

Table II shows that the proposed algorithm MRC-CCG obtains accurate solutions with low AE (see Appendix D for the classification errors on each dataset) and achieves around a 10x speedup for datasets with a large number of samples ('pulsar', 'house16', 'cats vs dogs', and 'yearbook'), and around a 2x speedup for datasets with a large number of samples and features ('rcv1', 'real_sim', and 'news20'). In addition, Table III shows that the algorithm enables the efficient learning of MRCs for multi-class settings and can achieve around a 100x speedup with increasing number of classes.

The experimental results show that the proposed algorithm MRC-CCG achieves faster convergence than existing methods with differences in the worst-case error probability lower than 1e-3. The algorithm enables to significantly improve the learning efficiency with large-scale data, especially for cases with large number of samples. In addition, the algorithm provides

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Table II. Performance assessment in terms of training time (in secs) and average error (AE) for binary classification.

Dataset	SVMs			MRCs				
	LP	CCG		LP	SUB		CCG	
	Time	Time	AE	Time	Time	AE	Time	AE
pulsar	1.1e+2 ± 1e+0	1.6e+2 ± 1e+1	5e-11	7.0e+1 ± 5e+0	2.5e+2 ± 7e+1	4e-3	7.2e+0 ± 1e+0	8e-5
house16	1.3e+2 ± 1e+0	1.7e+2 ± 3e+0	1e-11	9.4e+1 ± 1e+0	3.2e+2 ± 8e+1	3e-3	7.1e+0 ± 4e+0	9e-5
cats vs dogs	1.7e+2 ± 8e+0	1.0e+2 ± 9e+0	2e-11	2.0e+2 ± 2e+1	4.0e+2 ± 1e+2	6e-4	2.9e+1 ± 5e+0	1e-5
yearbook	2.8e+2 ± 1e+1	3.0e+2 ± 3e+1	1e-11	3.3e+2 ± 2e+1	6.2e+2 ± 1e+2	3e-4	3.8e+1 ± 5e+0	5e-6
rcv1	5.6e+3 ± 2e+2	2.5e+2 ± 2e+1	1e-4	2.1e+4 ± 1e+3	3.6e+4 ± 8e+3	6e-2	8.1e+3 ± 2e+3	2e-3
real_sim	9.4e+3 ± 2e+2	6.6e+3 ± 2e+3	4e-1	5.7e+4 ± 6e+3	6.2e+4 ± 2e+4	5e-2	1.9e+4 ± 8e+2	9e-4
news20	-	2.1e+4 ± 5e+3	-	-	-	-	1.4e+5 ± 8e+3	-

Table III. Training times (in secs) for multi-class classification.

Dataset	SVM-MULTI	MRC-SUB	MRC-CCG
satellite	3.5e+2 ± 6e+0	4.3e+2 ± 1e+2	7.4e+1 ± 5e+0
dry_bean	1.6e+2 ± 7e+0	2.5e+3 ± 1e+2	4.2e+1 ± 5e+0
optidigits	4.9e+2 ± 1e+1	5.8e+3 ± 4e+2	3.4e+2 ± 2e+1
mnist	9.2e+4 ± 5e+4	1.8e+5 ± 4e+4	2.5e+3 ± 3e+2
fashion_mnist	-	1.9e+5 ± 2e+4	2.9e+3 ± 1e+2
cifar10	-	1.2e+5 ± 2e+4	4.2e+3 ± 2e+2
cifar100	-	-	9.7e+4 ± 3e+3

improved scalability upto a sizeable number of classes.

V. Conclusion

In this paper, we presented an algorithmic framework based on the combination of constraint and column generation for the efficient learning of minimax risk classifiers (MRCs) in general large-scale scenarios. The iterative algorithm utilizes a greedy constraint selection approach at each iteration that results in a complexity that scales quasi-linearly with the number of classes. We present theoretical results that describe the convergence of the proposed algorithm for multiple scenarios of large-scale data. In particular, the results bound the difference in the worst-case error probability at each iteration. The numerical results assess the efficiency of the proposed algorithm in comparison with state-of-the-art methods using multiple binary and multi-class datasets. The results show that the proposed algorithm provides a significant efficiency increase for large-scale data. Moreover, the algorithm achieves a significantly reduced computational complexity for multi-class classification tasks (upto 100x speedup) and enables scalability to a sizeable number of classes.

Acknowledgement

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Appendix A
Proof of theorem 1

Proof. At each iteration, the algorithm adds and removes constraints to the primal LP (4), that is, variables to the dual LP (6). The inequality (10) follows by noting that the warm-start to the dual at iteration $k+1$ is a feasible solution with an objective value that equals the worst-case error probability R^k at iteration k. In particular, the warm-start is feasible at $k+1$ and has value R^k because it is obtained from the solution at iteration k by removing and adding features corresponding to zero coefficients. In the following, we prove the inequality (11).

The second inequality in (11) follows by noting that R^* is the minimum value corresponding with all the constraints and will be greater than or equal to any optimal value corresponding with a subset of constraints. The first inequality in (11) is obtained as follows.

Let $\boldsymbol{\mu}_1^{k_0}, \boldsymbol{\mu}_2^{k_0} \in \mathbb{R}^m$, $\nu^{k_0} \in \mathbb{R}$ be the primal solution at iteration k_0 for a subset of primal constraints $\mathcal{I}$, then we have

$$\mathbf{F}(\boldsymbol{\mu}_1^{k_0} - \boldsymbol{\mu}_2^{k_0}) - \nu^{k_0}\mathbf{1} \preceq \mathbf{b} + \hat{\epsilon}_1\mathbf{1} \tag{14}$$

since $\hat{\epsilon}_1$ is the largest violation in the constraints of the primal. Now, consider the following dual problem corresponding to the primal (4) with constraints as in (14).

$$\begin{aligned} \max_{\boldsymbol{\alpha}\in\mathbb{R}^p, \boldsymbol{\alpha}\succeq 0} \quad & -(\mathbf{b}+\hat{\epsilon}_1\mathbf{1})^{\mathrm{T}}\boldsymbol{\alpha} \\ \text{s.t.} \quad & \boldsymbol{\tau}-\boldsymbol{\lambda} \preceq \mathbf{F}^{\mathrm{T}}\boldsymbol{\alpha} \preceq \boldsymbol{\tau}+\boldsymbol{\lambda},\ \mathbf{1}^{\mathrm{T}}\boldsymbol{\alpha}=1. \end{aligned} \tag{15}$$

If R^* is the optimal solution of the MRC LP (4) using all the constraints, then the corresponding dual solution $\boldsymbol{\alpha}^*$ is also a feasible solution for the dual LP (15). Therefore, by weak duality ([Bertsimas and Tsitsiklis, 1997, Theorem 4.3]), we have that

$$\begin{aligned} -(\mathbf{b}+\hat{\epsilon}_1\mathbf{1})^{\mathrm{T}}\boldsymbol{\alpha}^* \leq\ & (\boldsymbol{\lambda}-\boldsymbol{\tau})^{\mathrm{T}}\boldsymbol{\mu}_1^{k_0} \\ & + (\boldsymbol{\tau}+\boldsymbol{\lambda})^{\mathrm{T}}\boldsymbol{\mu}_2^{k_0} + \nu^{k_0} = \mathrm{R}^{k_0} \end{aligned} \tag{16}$$

since $\boldsymbol{\mu}_1^{k_0}, \boldsymbol{\mu}_2^{k_0}$, and ν^{k_0} is a feasible solution for the primal LP corresponding with relaxed constraints in (14). Therefore, the first inequality in (11) is obtained because

$$\mathrm{R}^* - \hat{\epsilon}_1\mathbf{1}^{\mathrm{T}}\boldsymbol{\alpha}^* = \mathrm{R}^* - \hat{\epsilon}_1 \leq \mathrm{R}^{k_0} \tag{17}$$

since $\mathbf{1}^{\mathrm{T}}\boldsymbol{\alpha}^* = 1$. Hence, the first inequality of (11) holds for any $k \geq k_0$ since $\mathrm{R}^k \geq \mathrm{R}^{k_0}$ due to the monotonic increase of the worst-case error probability. □

Appendix B
Proof of theorem 2

Proof. The first inequality follows from Theorem 1, based on the maximum primal constraint violation $\hat{\epsilon}_1$ at iteration k.

We now prove the second inequality by applying weak duality [Bertsimas and Tsitsiklis, 1997]. Let $\boldsymbol{\mu}_1^k, \boldsymbol{\mu}_2^k \in \mathbb{R}^m$, $\nu^k \in \mathbb{R}$ be the primal solution, and $\boldsymbol{\alpha}^k \in \mathbb{R}^p$ the dual solution

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for selected constraints $\mathcal{I}$ and features $\mathcal{J}$. Since $\hat{\epsilon}_2$ is the maximum violation in the dual constraints, we have:

$$\boldsymbol{\tau} - \boldsymbol{\lambda} - \hat{\epsilon}_2 \mathbf{1} \preceq \mathbf{F}^{\mathrm{T}} \boldsymbol{\alpha}^k \preceq \boldsymbol{\tau} + \boldsymbol{\lambda} + \hat{\epsilon}_2 \mathbf{1} \tag{18}$$

Now, we consider the corresponding relaxed primal problem. The optimal MRC solution $(\boldsymbol{\mu}_1^*, \boldsymbol{\mu}_2^*, \nu^*)$ remains feasible under this relaxation. Then, by weak duality:

$$-\mathbf{b}^{\mathrm{T}} \boldsymbol{\alpha}^k \leq (\boldsymbol{\lambda} + \hat{\epsilon}_2 \mathbf{1} - \boldsymbol{\tau})^{\mathrm{T}} \boldsymbol{\mu}_1^* + (\boldsymbol{\tau} + \boldsymbol{\lambda} + \hat{\epsilon}_2 \mathbf{1})^{\mathrm{T}} \boldsymbol{\mu}_2^* + \nu^*$$

which implies $\mathrm{R}^k \leq \mathrm{R}^* + \hat{\epsilon}_2 \|\boldsymbol{\mu}^*\|$ since $\boldsymbol{\mu}_1^* = (\boldsymbol{\mu}^*)_+$ and $\boldsymbol{\mu}_2^* = (-\boldsymbol{\mu}^*)_+$. □

APPENDIX C
HYPER-PARAMETER TUNING

In the following, we present results that analyze the effect of the hyper-parameters ϵ_1, ϵ_2, $n_{\max}$, and $m_{\max}$ on the efficiency of Algorithm 1.

A. Effect of ϵ_1 and ϵ_2

Hyper-parameters ϵ_1 and ϵ_2 determine the threshold for the primal and dual constraints' violations, that is, the proposed algorithm only selects constraints that have a higher violation than the threshold in each iteration. Therefore, ϵ_1 and ϵ_2 influence the total number of constraints and features selected, and thereby, the overall time taken by the algorithm. Moreover, Theorem 2 shows that ϵ_1 has direct effect on the accuracy of the solution while the effect of ϵ_2 depends on the sparsity in the coefficient $\boldsymbol{\mu}^*$. Table IV shows the trade-off between the time and accuracy of the algorithm for different values of the hyper-parameters ϵ_1 and ϵ_2. The table presents the average relative error (ARE), the classification error, and the training time obtained by the algorithm using the datasets "rcv1" and "real_sim" with large number of samples and features. The results are obtained for the hyper-parameters $n_{\max} = m_{\max} = 400$. We observe that smaller values of the hyper-parameter can obtain very accurate solutions while taking more time. On average, a good trade-off between accuracy and time is obtained for ϵ_1 =1e-2 and ϵ_2 =1e-5.

Table IV. Performance assessment for different values of ϵ_1 and ϵ_2.

ϵ_1	ϵ_2	Dataset					
		rcv1			real_sim		
		Time	Error	AE	Time	Error	AE
1e-2	1e-3	9.8e+1	.07	8.5e-2	2.7e+2	.24	4.7e-2
1e-2	1e-4	3.6e+3	.05	1.0e-2	5.6e+3	.16	1.0e-2
1e-2	**1e-5**	**9.6e+3**	**.04**	**2.0e-3**	**2.0e+4**	**.13**	**9.2e-4**
1e-3	1e-3	9.4e+1	.07	8.5e-2	3.1e+2	.24	4.7e-2
1e-3	1e-4	3.0e+3	.05	2.1e-2	6.5e+3	.16	1.1e-2
1e-3	1e-5	1.2e+4	.04	2.2e-3	2.5e+4	.13	9.1e-4
1e-4	1e-3	1.0e+2	.07	8.4e-2	3.6e+2	.24	4.6e-2
1e-4	1e-4	3.3e+3	.05	2.0e-2	8.8e+3	.16	1.1e-2
1e-4	1e-5	8.8e+3	.04	2.2e-3	2.9e+4	.13	9.3e-4

B. Effect of n_{max} and m_{max}

Hyper-parameters $n_{\max}$ and $m_{\max}$ determine the maximum number of constraints and features selected by the proposed algorithm in each iteration. Therefore, $n_{\max}$ and $m_{\max}$ have an effect on the complexity per iteration and the total number of iterations of the proposed algorithm. Particularly, increasing the value of $n_{\max}$ or $m_{\max}$ decreases the number of iterations for convergence at the expense of increasing the complexity per iteration. In Figure 3, we present results that show the effect of $n_{\max}$ and $m_{\max}$ on the overall computational complexity of the algorithm on datasets 'real_sim' and 'rcv1'. We set ϵ_1 =1e-2 and ϵ_2 =1e-5 to obtain accurate results. In practice, we observe that $300 < n_{\max} = m_{\max} < 1500$ obtains a good compromise between the number of iterations required for convergence and the complexity per iteration. All the numerical results in the main paper are obtained for $n_{\max} = m_{\max} = 400$.

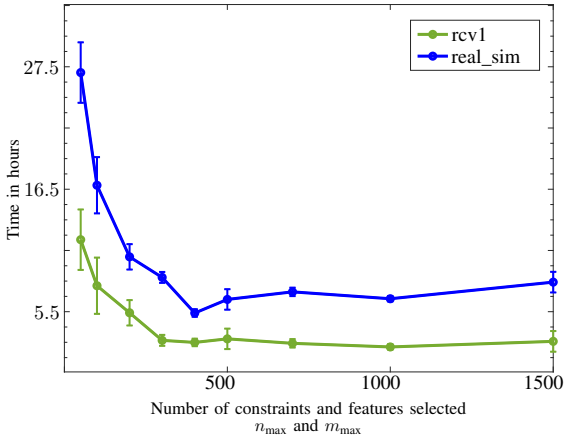


Figure 3. Effect of hyper-parameters $n_{\max}$ and $m_{\max}$ on training time.

APPENDIX D
CLASSIFICATION ERRORS ALONG WITH TRAINING TIMES ON MULTIPLE BENCHMARK DATASETS

In this section, we present results to show that the proposed algorithm MRC-CCG enables efficient learning while achieving classification errors similar to the state-of-the-art. Table V presents results corresponding to binary classification datasets while Table VI presents results corresponding to multi-class classification datasets. Moreover, the results show that the proposed algorithm achieves competitive classification errors compared to the methods based on SVMs. In addition, the results present the worst-case error error probability obtained by the methods for MRCs that can serve to provide an upper bound on the classification error.

REFERENCES

Zeynep Akata, Florent Perronnin, Zaid Harchaoui, and Cordelia Schmid. Good practice in large-scale learning for image classification. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 36(3), 2013.

Samy Bengio, Jason Weston, and David Grangier. Label embedding trees for large multi-class tasks. In *Advances in Neural Information Processing Systems*, volume 23, 2010.

Dimitris Bertsimas and John N. Tsitsiklis. *Introduction to linear optimization*, volume 6. Athena Scientific, Belmont, MA, 1997.

Kartheek Bondugula, Santiago Mazuelas, and Aritz Pérez. Efficient learning of minimax risk classifiers in high dimensions. In *Conference on Uncertainty in Artificial Intelligence*, 2023.

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Table V. Training time (in secs) and classification error (CE) for binary classification.

Dataset	SVMs				MRCs								
	LP		CCG		LP			SUB			CCG		
	Time	CE	Time	CE	Time	CE	R^*	Time	CE	R^{SUB}	Time	CE	R^{CCG}
pulsar	1.1e+2	.02	1.6e+2	.02	7.0e+1	.04	.06	2.5e+2	.05	.06	7.2e+0	.04	.06
house16	1.3e+2	.25	1.7e+2	.25	9.4e+1	.26	.27	3.2e+2	.27	.28	7.1e+0	.26	.27
cats vs dogs	1.7e+2	.02	1.0e+2	.02	2.0e+2	.01	.21	4.0e+2	.01	.21	2.9e+1	.01	.21
yearbook	2.8e+2	.07	3.0e+2	.07	3.3e+2	.11	.31	6.2e+2	.11	.31	3.8e+1	.11	.31
rcv1	5.6e+3	.03	2.5e+2	.03	2.1e+4	.04	.19	3.6e+4	.05	.25	8.1e+3	.04	.19
real_sim	9.4e+3	.04	6.6e+3	.04	5.7e+4	.12	.20	6.2e+4	.25	.25	1.9e+4	.13	.20
news20	-	-	2.1e+4	.07	-	-	-	-	-	-	1.4e+5	.04	.17

Table VI. Training times (in secs) and classification error (CE) for multi-class classification.

Dataset	SVM-MULTI		MRC-SUB			MRC-CCG		
	Time	CE	Time	CE	R^{SUB}	Time	CE	R^{CCG}
satellite	3.5e+2	.12	4.3e+2	.17	.32	7.4e+1	.15	.26
dry_bean	1.6e+2	.44	2.5e+3	.57	.57	4.2e+1	.40	.49
optdigits	4.9e+2	.03	5.8e+3	.06	.37	3.4e+2	.02	.22
mnist	9.2e+4	.06	1.8e+5	.05	.44	2.5e+3	.05	.41
fashion	-	-	1.9e+5	.90	.90	2.9e+3	.17	.46
cifar_10	-	-	1.2e+5	.90	.90	4.2e+3	.17	.57
cifar_100	-	-	-	-	-	9.7e+4	.55	.66

Kartheek Bondugula, Verónica Álvarez, Jose I. Segovia-Martín, Aritz Pérez, and Santiago Mazuelas. MRCpy: A library for minimax risk classifiers. *arXiv preprint*, arXiv:2108.01952, 2024.

Léon Bottou. Large-scale machine learning with stochastic gradient descent. In *19th International Conference on Computational Statistics*, 2010.

Léon Bottou and Olivier Bousquet. The tradeoffs of large scale learning. In *Advances in Neural Information Processing Systems*, volume 20, 2007.

Michael Celentano and Andrea Montanari. Fundamental barriers to high-dimensional regression with convex penalties. *The Annals of Statistics*, 50(1), 2022.

Antoine Dedieu, Rahul Mazumder, and Haoyue Wang. Solving L1-regularized SVMs and related linear programs: Revisiting the effectiveness of column and constraint generation. *Journal of Machine Learning Research*, 23, 2022.

Jacques Desrosiers and Marco E. Lübbecke. *A primer in column generation*. Springer, 2005.

Rizal Fathony, Anqi Liu, Kaiser Asif, and Brian Ziebart. Adversarial multiclass classification: A risk minimization perspective. In *Advances in Neural Information Processing Systems*, volume 29, 2016.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016.

Kenji Kawaguchi and Haihao Lu. Ordered SGD: A new stochastic optimization framework for empirical risk minimization. In *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics*, 2020.

Li-Jia Li, Hao Su, Li Fei-Fei, and Eric Xing. Object bank: A high-level image representation for scene classification & semantic feature sparsification. In *Advances in Neural Information Processing Systems*, volume 23, 2010.

Santiago Mazuelas, Andrea Zanoni, and Aritz Pérez. Minimax classification with 0-1 loss and performance guarantees. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

Santiago Mazuelas, Yuan Shen, and Aritz Pérez. Generalized maximum entropy for supervised classification. *IEEE Transactions on Information Theory*, 68, 2022.

Santiago Mazuelas, Mauricio Romero, and Peter Grunwald. Minimax risk classifiers with 0-1 loss. *Journal of Machine Learning Research*, 24(208), 2023.

Francesca Mignacco, Florent Krzakala, Yue Lu, Pierfrancesco Urbani, and Lenka Zdeborova. The role of regularization in classification of high-dimensional noisy gaussian mixture. In *Proceedings of the 37th International Conference on Machine Learning*, 2020.

Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar. *Foundations of machine learning*. MIT press, Cambridge, MA, second edition, 2018.

Ali Rahimi and Benjamin Recht. Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems*, volume 20, 2008.

Shai Shalev-Shwartz, Yoram Singer, and Nathan Srebro. Pegasos: Primal estimated sub-gradient solver for SVM. In *Proceedings of the 24th International Conference on Machine Learning*, 2007.

Yoram Singer and John C Duchi. Efficient learning using forward-backward splitting. In *Advances in Neural Information Processing Systems*, volume 22, 2009.

Mingkui Tan, Li Wang, and Ivor W Tsang. Learning sparse SVMs for feature selection on very high dimensional datasets. In *Proceedings of the 27th International Conference on Machine Learning*, 2010.

Lifeng Wang, Xiaotong Shen, and Yuan F Zheng. On L1-norm multi-class support vector machines. In *International Conference on Machine Learning and Applications*, volume 5. IEEE, 2006.

Meng Wang, Weijie Fu, Xiangnan He, Shijie Hao, and Xindong Wu. A survey on large-scale machine learning. *IEEE Transactions on Knowledge and Data Engineering*, 34(6), 2020.

Guo-Xun Yuan, Chia-Hua Ho, and Chih-Jen Lin. Recent advances of large-scale linear classification. *Proceedings of the IEEE*, 100, 2012.

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